

Exam Time: 90 minutes
SHOW YOUR DETAILED SOLUTIONS AND CALCULATIONS. Final answers will be marked only at (10 marks)

Answer the following questions

Q1- Fit a straight-line to the data provided in the following Table:

x_i	5	7	9	11	13
y_i	6	4.5	3.4	2.3	1.4

The equation of the line is:

The value of coefficient of determination, r^2 is:

Q2- Compute the integration: $\int_2^4 \frac{1}{(x-1)^2} dx$, using Work to five significant figures.

Apply Romberg integration taking $n=1, n=2$ & $n=4$ up to $R(3, 2)$.

Answer:

Q3- Evaluate the following double integration: $\int_1^3 \int_1^3 (2x + y) dx dy$.

Take $dx = 0.5$ in the x -direction, $dy = 0.5$ in the y -direction. Apply the Trapezoidal rule in the x -direction and in the y -direction.

Answer:

Q4- Maximize the function: $f(x, y) = xy + 4x - x^2 - y^2 - 4$

Apply the Optimal steepest ascent method, use point $(2, 2)$ as the initial estimate of the optimal solution. The optimum point after conducting one iteration is:

Answer:

Q5- Consider the ordinary differential equation: $\frac{dy}{dx} = x^2 - 2y^2 + x$

Given: $x_0 = 1, y_0 = 2, h = 0.5$

Use the Runge-Kutta method to the 4th order to find $y(1.5)$.

Answer:

End of Questions

x_i $f(x)$

-1 3.137

0 6.132

1 4.311

2 7.525

3 2.289

(a)

$$I = \frac{1}{3} (3.137 + 2.289 + 4(6.132 + 7.525) + 2(4.311))$$

$$I = 22.892$$

(b) Romberg method

$$R(1,0) \rightarrow h = 4$$

$$f(-1) = 3.137$$

$$f(3) = 2.289$$

$$R(1,0) = \frac{4}{2} (3.137 + 2.289)$$

$$= 10.852$$

$$R(2,0) = \frac{2}{2} (3.137 + 2.289 + 2(4.311))$$

$$h = 2$$

$$= 14.048$$

$$R(3,0) = \frac{1}{2} (3.137 + 2.289 + 2(6.132 + 4.311 + 7.525))$$

$$h = 1$$

$$R(3,0) = 20.681$$

$$R(2,1) = \frac{14 R(2,0) - R(1,0)}{3} = 15.113$$

$$R(3,1) = \frac{14 R(3,0) - R(2,0)}{3} = 22.892$$

$$R(3,2) = \frac{16 R(3,1) - R(2,1)}{15} = 23.4106$$

$$\begin{bmatrix} 2 & -1 & -1 & 2 \\ 1 & -2 & 1 & -4 \\ 1 & 1 & \lambda & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 1 & -4 \\ 2 & -1 & -1 & 2 \\ 1 & 1 & \lambda & 4 \end{bmatrix} \begin{array}{l} -2R_1 + R_2 \\ -1R_1 + R_3 \end{array}$$

$$\begin{bmatrix} 1 & -2 & 1 & -4 \\ 0 & 1 & -1 & \frac{10}{3} \\ 0 & 3 & \lambda-1 & 8 \end{bmatrix} \xrightarrow{-3R_2 + R_3} \begin{bmatrix} 1 & -2 & 1 & -4 \\ 0 & 3 & -3 & 10 \\ 0 & 3 & \lambda-1 & 8 \end{bmatrix} \div 3$$

$$\begin{bmatrix} 1 & -2 & 1 & -4 \\ 0 & 1 & -1 & \frac{10}{3} \\ 0 & 0 & \lambda+2 & -2 \end{bmatrix}$$

$$\lambda+2 = -2 \Rightarrow \text{no. sol. ell. lines}$$

$$\boxed{\lambda = -2}$$

$$\text{Q2} \quad \frac{dy}{dx} = \frac{-y \cos(y)}{2}$$

$$y_0 = \pi \quad x_0 = \pi$$

$$y\left(\frac{3\pi}{2}\right) = ?$$

$$h = \frac{3\pi}{2} - \pi$$

$$\boxed{h = \frac{\pi}{2}}$$

$$\boxed{\frac{2\pi}{2} + \frac{\pi}{2} = \frac{3\pi}{2}}$$

$$\frac{4\pi}{4} + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$k_1 = \frac{\pi}{2} f(\pi, \pi) = 2.467$$

$$k_2 = \frac{\pi}{2} f\left(\frac{3\pi}{2}, 4.3751\right) = 1.137$$

$$k_3 = \frac{\pi}{2} f\left(\frac{3\pi}{2}, 3.7101\right) = 2.4556$$

$$k_4 = \frac{\pi}{2} f\left(\frac{3\pi}{2}, 5.5972\right) = \text{error} - 3.4016$$

$$y\left(\frac{3\pi}{2}\right) = y(\pi) + \frac{1}{6} \left(2.467 + 2(2.4556) + 2(1.137) + (-3.4016) \right)$$

$$\boxed{y\left(\frac{3\pi}{2}\right) = 3.1676}$$

$$\boxed{y = 4.1834}$$

Q1. $\lambda = -2$

Q2. $y\left(\frac{3\pi}{2}\right) = 4.18325$

Q3. $2.891x^3 + 1.208x^2$
 $+ 2.109x + 1$

Q4. Simpson's $\frac{1}{3}$:
 $A = 22.892$

Romberg's :

$A = 23.4108.$